Ishaan Somya Mehrotra

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EDUCATION

Duke University, Durham, NC **August 2021 – May 2025 | August 2026 – May 2027**

4+1 M.S.E. in Biomedical Engineering (Expected: **May 2027**) | B.S.E. in Biomedical Engineering | B.S. in Computer Science | Minor in Chemistry

Relevant Coursework: Data Science, Databases, Data Structures, Algorithms, Business Fundamentals for Engineers, Differential Equations, Generative AI in Protein Design, Customer Empathy & Brand Design, Innovation & Entrepreneurship, Quantitative Physiology, Biostatistics, Medical Design Prototyping, Cellular & Molecular Engineering, Drug Delivery, Biotech Design, Asset Pricing & Risk Management, AI in BME, Management of High Tech Industries

Skills: Excel (Advanced), PowerPoint, Sourcing, Market Research, Financial Modeling, Valuation, Due Diligence, SQL, R, Python, Power BI, Tableau, LLMs, Bloomberg, A/B Testing, PitchBook, GCP | **Programs:** RA Capital Management NPLB Fellow, Duke Investor Training Program, Duke I&E VC Foundations

WORK EXPERIENCE

Shiftwave San Francisco, CA (Remote)

AI & GTM Intern | Nervous System Recovery & Regulation Device Medical Technology Startup August 2026 – Present

- Conducting customer discovery, market sizing, and unit economics analysis across soccer, golf, and Formula 1 to recommend priorities for GTM expansion to leadership; built an AI-assisted pipeline scoring and ranking target orgs on fit and deployment readiness informing business development

1824 Health Ventures Charleston, SC (Remote)

Venture Capital Intern | Medical University of South Carolina Innovation Program May 2026 – August 2026

- Supported diligence on seed-stage biotech startups across trial readouts, regulatory milestones, IP, and financing needs; observed manual portfolio monitoring processes and built AI-enabled tracker pulling data, filing, and financing events into one holistic view, replacing ad-hoc check-ins for the investment team; pressure-tested founder, product, and market-fit risk into investment recommendations, then codified these risks as tracker criteria

CVS Health Chicago, IL (Remote)

Business Analyst | Oak Street Health, Population Health Division | Data, Strategy, & Operations | Fortune 10 August 2025 – Present

- Developed executive dashboards integrating clinical, payer claims, and financial data across 250k+ patients; quantified $132M+ annual respiratory cost exposure and $171 net savings per vaccine to set the business case and coverage targets for the 2026-27 OSH vaccine campaign, executed via flu standing orders and Epic gap prompts; supported org-wide Google Gemini adoption through provider onboarding and implementation walkthroughs
- Ranked 26 payer-markets by Stars penalty exposure to focus on the top 3 regions, then sized CBP and A1c gaps into a per-clinic run rate (~4 closures/week/center) in PowerQuery and BigQuery; found 264 unscheduled patients owned home BP cuffs, making gaps phone-closable, and built the 4-week pilot (targeted patient tier segmentation routing to MA/PharmD/RN; readings logged to discrete Epic EHR fields) driving a 2.6x YoY lift in BP rechecks

Lipo-ImmunoTech Charleston, SC (Remote)

Strategy & Data Analytics Intern | Phase I/II Biotechnology Startup January 2025 – July 2025

- Evaluated commercialization strategies and clinical development pathways for a Phase I/II ceramide-based cancer immunotherapy platform by synthesizing preclinical efficacy data, regulatory feasibility, and market dynamics to assess growth opportunities and clinical translation potential
- Conducted market landscape/competitive analyses (Excel, Power BI) across global immuno-oncology and cell therapy pipelines, identifying projected >$500M TAM opportunities; informed business development and client engagement methods through standard-of-care research and financial modeling
- Collaborated cross-functionally with biopharmaceutical executives, clinicians, and scientists to support fundraising, cap table review, and strategic partnership initiatives tied to NCI/STTR-funded therapeutic programs and SCbioDrive venture acceleration activities; translated cGMP cell-manufacturing workflows and oncology data into scientific value propositions for investors and partners driving early-phase clinical trial design

Programmable Biology Group | Chatterjee Lab Durham, NC

Translational Research Intern | Experimental Validation of AI-designed Peptides for Targeted Protein Modulation August 2024 – May 2025

- Validated AI-designed peptide binders using transformer-based protein modeling tools (PepMLM, moPPIt v3) through experimental assays to assess binding affinity and optimize translational viability for proteome editing applications, contributing to product development strategy and early-stage go/no-go decisions; supported evaluation of strong commercial potential for ~70 outputted candidates through benchmarking and a comparative analysis
- Leveraged data-driven problem solving (SQL, R), analytical thinking, and process optimization techniques to refine design strategy, driving a 40% increase in high-affinity binders, enhancing early-stage drug development efficiency and streamlining pipeline workflows for peptide-based therapeutics

Bass Connections: Development of a VR & Haptic Feedback System for Interventional (IR) Training Durham, NC

Product Lead | Duke University School of Medicine August 2024 – June 2025

- Orchestrated development of a high-fidelity VR simulation platform for IR training at Duke Health by integrating haptic hardware, real-time force sensing, and Unity-based environments; coordinated a 6-member team to collect, analyze, and translate spatial and force-sensor data through VIVE OpenXR, Raspberry Pi, and WebSocket-enabled pipelines, earning Best Interactive Display at the 2025 Fortin Foundation Bass Connections showcase

Duke Consulting Club: Gamifying Risk Identification for Alcohol Use Behaviors Across Countries and Cultures Durham, NC

Project Manager | Duke GEMINI Research Center August 2023 – June 2024

- Owned end-to-end product strategy and shipped deployment of Flutter-based SaaS clinical decision tool for Alcohol Use Disorder (AUD) assessment across 3 international markets and 5 languages; leveraged psychometric modeling and behavioral analytics to improve risk stratification accuracy by 85%, driving a 30% increase in AUD identification; engaged healthcare stakeholders through 1:1 interviews to scale API-integrated health applications

CharlestonKix Charleston, SC

Founder & CEO December 2017 – August 2023

- Founded a youth sports nonprofit, raising $6k and securing 10 sponsors to expand soccer access for 32 low-income youth through 100+ lessons

PROJECTS

ProtMistral-Tiny: 10,000x Compute-Efficient Protein Language Models (pLMs) Development & Evaluation Durham, NC

AI & ML Analyst | Team of 5 January 2025 – May 2025

- Trained a 15k-parameter Mistral pLM matching a 10,000x larger baseline on AlphaFold structure quality in 3 A100 GPU hours, cutting compute cost
- Fine-tuned ESM-2 on dense DMS fitness labels (PyTorch, HuggingFace) & built the eval suite (pLDDT/pTM/PAE, Shannon Entropy, k-mer diversity) that caught a tradeoff the headline metric hid: thermostability R^2 fell 0.41 to 0.28 while generated-structure pLDDT rose 48 to 79 at matched diversity

Undergraduate Research Project: Open Design Studio | Duke Social Science Research Institute Durham, NC

Product & User Research Analyst | Community Health Innovation Fellowship July 2022 – June 2023

- Conducted 30+ user interviews with students, patients, clinicians, and public health leaders to define unmet user needs for a solution connecting healthcare nonprofits with technical talent in the Durham community; translated key findings into a product requirements document (PRD) and iteratively tested Figma prototypes; co-developed a for-credit course engaging 100+ students and generating up to 15 health-policy project proposals

Duke's 501(c)(3) Chapter of Healthy Hands Initiative Durham, NC

Executive Board | Vice President of Strategy & Partnerships February 2022 – May 2023

- Drove growth strategy and secured 15+ external partnerships for a clean water healthcare venture addressing preventable disease in Northeast India
- Raised $50k in competitive non-dilutive capital through grant financing; designed a hybrid public-private funding model for long-term sustainability